

PWM-Based Double Integral Sliding Mode Control for Current Control Loop of DC-DC Boost Converter

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Abstract. The use of dc-dc boost converters is common for increasing output voltage and supplying power. These converters, being time-varying nonlinear circuits, encounter uncertainties in load and variations in input voltage and current over a wide range. Nonlinear sliding mode control (SMC) has emerged as a promising strategy, offering advantages in terms of stability and robustness against parameter and load fluctuations. To address switching frequency variations in conventional hysteresis-based SMC, the indirect SMC or pulse-width modulation (PWM)-based SMC has been adopted. However, the adoption of PWM-based SMC raised concerns regarding steady-state error in power converters. Despite incorporating the integral of the state variable into the sliding surface of PWM-based SMC, the steady-state error still exists. Focusing on the current control loop of a dc-dc boost converter with the goal of reducing steady-state error, this study proposes double integral sliding mode control (DISMC), which employs a sliding surface that includes the double integral term of the current error. Simulation and experimental results demonstrate that the proposed approach effectively mitigates the steady-state error of the inductor current.

Keywords. sliding mode control (SMC), double integral SMC (DISMC), pulse width modulation (PWM), boost converter, current control

1. Introduction

Power electronic converters have become integral to everyday life and various industrial applications, including LED lighting systems, battery energy storage systems, and photovoltaic systems [1–3]. In cases where there is a low-level input voltage source, like a battery, a dc-dc boost converter is typically utilized to increase the output voltage and power the load. For a dc bus system with a battery, it is crucial to pay attention to the performance of the current control loop because charging and discharging current signif-

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icantly impact the health of the battery, and the current supplied into the dc bus affects the dc bus voltage. The dc-dc boost converter is a time-varying nonlinear circuit, and its operation may encounter uncertainties in load and variations in input voltage and current across a wide range. Consequently, designing control loops that incorporate elements like robustness, stability, tracking, and regulation performance can be challenging.

The sliding mode controller (SMC) is a nonlinear controller gaining significant attention for enhancing the control performance of power electronic converters. It offers substantial advantages in terms of stability and robustness against parameter and load variations. However, two major concerns arise when using conventional hysteresis-based SMC in power converters, namely, the existence of steady-state error and inconstant switching frequency. The issue of steady-state error in hysteresis-based SMC can effectively be mitigated by constructing the sliding surface with the integral of the state variable, namely the integral SMC (ISMC). In regard to the second concern, the inherent unsteady switching frequency and high switching frequency of the SMC for converters may cause anxiety in filter design. Maintaining a constant switching frequency can be achieved by implementing indirect SMC or pulse-width modulation (PWM)-based SMC. This type of SMC has garnered significant attention and has been widely used. In [4], digital PWM-based sliding mode current control was employed for switched reluctance machine drives. A sliding surface is defined where a current reference tracking error and the first-order derivative of the current reference tracking error are chosen as state variables. In [5], to improve the power factor, a single-ended primary-inductor converter was controlled using PWM-based SMC, which includes an outer loop for output voltage control and an inner loop for inductor current control. In [6], the current control loop of the single-phase grid-connected inverter was regulated using PWM-based SMC, where the sliding surface incorporates both the inductor current error and the integral of the inductor current error. However, when PWM-based SMC is adopted, even though there is an integral of the state variable in the sliding surface, the steady-state error persists. In [7], for voltage mode control of dc-dc power converters, a sliding surface comprised of an additional double integral term of the state variable was introduced. The study demonstrated that the steady-state error can be eliminated by constructing the sliding surface with the double integral of the state variable. The ability to suppress time-varying disturbances and steady-state oscillations of the output voltage was also discussed in [8]. Mitigating steady-state error in PWM-based SMCs remains a challenging task.

This study focuses on the inductor current control loop of a dc-dc boost converter. In an effort to reduce steady-state error, the study proposes the adoption of PWM-based SMC with a sliding surface comprised of the double integral term of the error current, i.e., the double integral SMC (DISMC), for the inductor current control loop. The remainder of the paper is organized as follows: Section II describes the mathematical model of the dc-dc boost converter based on the continuous conduction mode. In Section III, both integral and double integral PWM-based SMCs are presented. The proposed control scheme is then validated with simulation and experimental results in Section IV. Finally, Section V concludes this study.

2. Mathematical Model of DC-DC Boost Converter

A dc-dc boost converter, as illustrated in Fig. 1, is analyzed based on its operation in continuous conduction mode. Its model is established using the control function, where

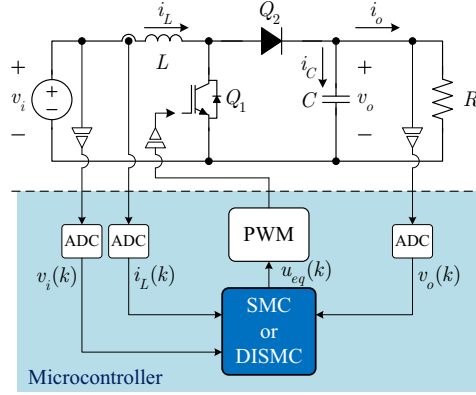


Figure 1. Dc-dc boost converter with the proposed control scheme.

u equals 1 when Q_1 is turned on, and u equals 0 when Q_1 is turned off. By defining $\bar{u}$ as $1 - u$, the mathematical model of the dc-dc boost converter can be expressed as follows:

$$\frac{di_L}{dt} = \frac{v_i}{L} - \frac{v_o \bar{u}}{L} \quad (1)$$

$$\frac{dv_o}{dt} = -\frac{i_o}{C} + \frac{i_L \bar{u}}{C} \quad (2)$$

Referring to (1) and (2), these are utilized in the development of current loop control techniques and the execution of computer simulations.

3. PWM-Based SMC for Current Control Loop

In this section, we describe the PWM-based SMCs used for regulating the inductor current to track its reference. To provide a clear explanation, we divide this section into two parts, namely ISMC and DISMC, as follows.

3.1. ISMC

For ISMC, an additional integral term of a state variable is considered in the control computation with the expectation of reducing steady-state error. The integral of the current error and the current error itself are chosen as state variables, as follows:

$$x_1 = \int i_{ref} - i_L dt \quad (3)$$

$$x_2 = \dot{x}_1 = i_{ref} - i_L$$

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = -\frac{v_i}{L} + \frac{v_o \bar{u}}{L} \quad (4)$$

The sliding surface is defined as:

$$S = \alpha_1 x_1 + \alpha_2 x_2 \quad (5)$$

where α_1 and α_2 represent sliding coefficients. By setting the derivative of S equal to zero and subsequently solving for u , we can derive the equivalent control function as follows:

$$\begin{aligned} \bar{u}_{eq} &= -\frac{1}{v_o} \left(L \frac{\alpha_1}{\alpha_2} x_2 - v_i \right) \\ &= -\frac{1}{v_o} \left(K_p x_2 - v_i \right) \end{aligned} \quad (6)$$

where K_p is defined as $L(\alpha_1/\alpha_2)$. As can be seen in (5) - (6), although there is an integral term of the state variable in the sliding function, the integral term of the error does not appear in the computation process of the equivalent control function.

3.2. DISMC

For DISMC, the controlled state variables are chosen with the expectation of incorporating the integral term of the error into the equivalent control function. These variables comprise the double integral of the current error, the integral of the current error, and the current error itself, expressed as:

$$\begin{aligned} x_1 &= \iint i_{ref} - i_L dt dt \\ x_2 &= \dot{x}_1 = \int i_{ref} - i_L dt \\ x_3 &= \dot{x}_2 = i_{ref} - i_L \end{aligned} \quad (7)$$

$$\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= x_3 \\ \dot{x}_3 &= -\frac{v_i}{L} + \frac{v_o \bar{u}}{L} \end{aligned} \quad (8)$$

Consequently, the sliding surface is represented as a linear combination of the three state variables, which takes the form:

$$S = \alpha_1 x_1 + \alpha_2 x_2 + \alpha_3 x_3 \quad (9)$$

where α_1 , α_2 , and α_3 represent sliding coefficients. Performing the same procedure as before, by setting the derivative of (9) equal to zero and subsequently solving for u , the equivalent control function of DISMC can be derived as follows:

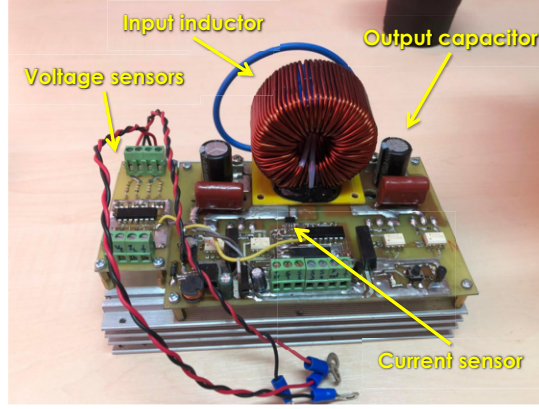


Figure 2. Experimental hardware for a dc-dc boost converter.

Table 1. Parameters of DC-DC Boost Converter

PARAMETERS	VALUE	UNIT
Input voltage v_i	12	V
Input inductance L	2	mH
Output capacitance C	470	μF
Load resistance R_o	30	Ω
Switching frequency f_{sw}	20	kHz
Sampling time T_s	50	μs

$$\begin{aligned}\bar{u}_{eq} &= -\frac{1}{v_o} \left(L \frac{\alpha_1}{\alpha_3} x_2 + L \frac{\alpha_2}{\alpha_3} x_3 - v_i \right) \\ &= -\frac{1}{v_o} \left(K_i x_2 + K_p x_3 - v_i \right)\end{aligned}\quad (10)$$

where K_i and K_p are defined as $L(\alpha_1/\alpha_3)$ and $L(\alpha_2/\alpha_3)$, respectively. To discretize the equivalent control function for implementation via a microcontroller, we utilize the trapezoidal rule to compute integrals, as follows:

$$x_2(k) = x_2(k-1) + \frac{T_s}{2} \left(x_3(k) + x_3(k-1) \right) \quad (11)$$

Observing in (10), the integral of the current error is included in the computation of the equivalent control function of DISMC, which is a significant factor in alleviating the steady-state error of the inductor current, as confirmed in the next section.

4. Simulation and Experimental Results

To verify the hypothesis and the proposed control scheme, simulations were conducted using MATLAB/Simulink. Additionally, a laboratory-built prototype shown in Fig. 2 was employed. The parameters are presented in Table I. The control system was implemented on the STM32F407VG central processing unit. Initially, ISMC was used for the current control loop of a dc-dc boost converter. The reference current was set to 1 A at

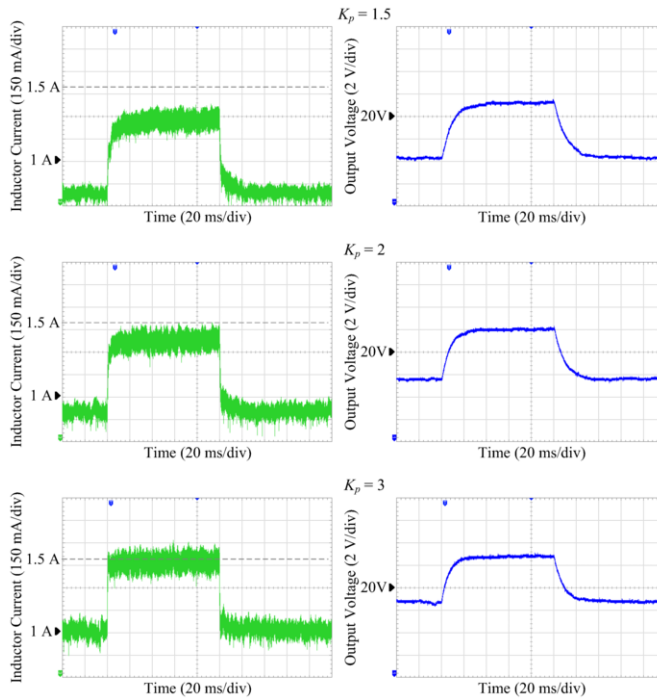


Figure 3. Inductor current and output voltage for ISMC in case of $K_p = 1.5, 2$, and 3 , respectively.

startup. After reaching a steady state, the reference current was changed to 1.5 A. The experiment was repeated by changing the K_p values to 1.5, 2, and 3, respectively. As depicted in Fig. 3, the inductor current changed according to the reference current, and the output voltage increased with the inductor current. Despite the decrease in steady-state error in the inductor current as K_p increased, some steady-state error still remained. Next, DISMC was employed for the current control loop. Following a similar procedure, the reference current was changed from 1 A to 1.5 A; after 100 ms, it was changed from 1.5 A to 1 A. With K_i fixed at 50, the experiment was repeated by changing the K_p values to 0.5, 1, and 2, respectively. As expected, Fig. 4 indicates that the steady-state error in the inductor current was reduced. In the case of $K_p = 0.5$, it can be observed that the current error value converges to zero before a step change occurs in the reference current. For the cases where $K_p = 1$ and $K_p = 2$, although the integral of the current error value did not reach zero, the steady-state error converged to zero. Then, with K_p fixed at 1, the experiment was repeated by changing the K_i values to 100 and 500, respectively. As depicted in Fig. 5, when K_i was increased, the overshoot increased, but the performance of steady-state error reduction improved noticeably faster. For DISMC, the integral of current error and current error were plotted in the phase plane. Although there were changes in the reference current and output voltage, the current error x_3 were always driven toward zero, as shown in Fig. 4 and Fig. 5. Despite slight discrepancies between the experimental and simulation results due to the use of an ideal boost converter model (excluding internal resistance in inductors and capacitors) in the simulation, the experimental results were consistent with the simulation results.

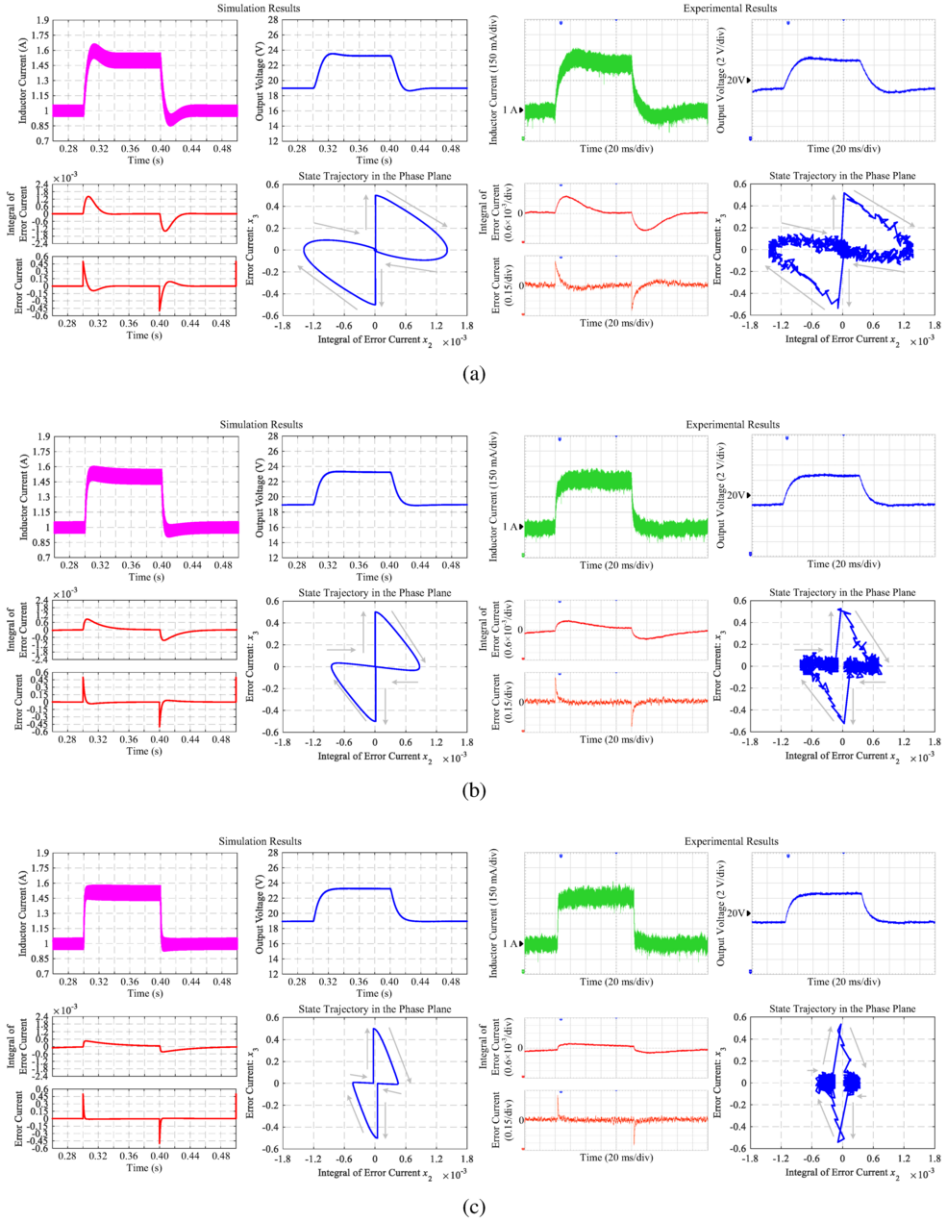


Figure 4. Inductor current, output voltage, current error, and the integral of current error for DISMC in case of $K_i = 50$ with (a) $K_p = 0.5$, (b) $K_p = 1$, and (c) $K_p = 2$.

5. Conclusion

In an effort to mitigate tracking errors in the inductor current of a dc-dc boost converter, this study proposes a DISMC approach for the current control loop. The sliding function of DISMC is constructed with the double integral term of the current error, along with the integral of the current error and the current error itself as state variables. Due to its

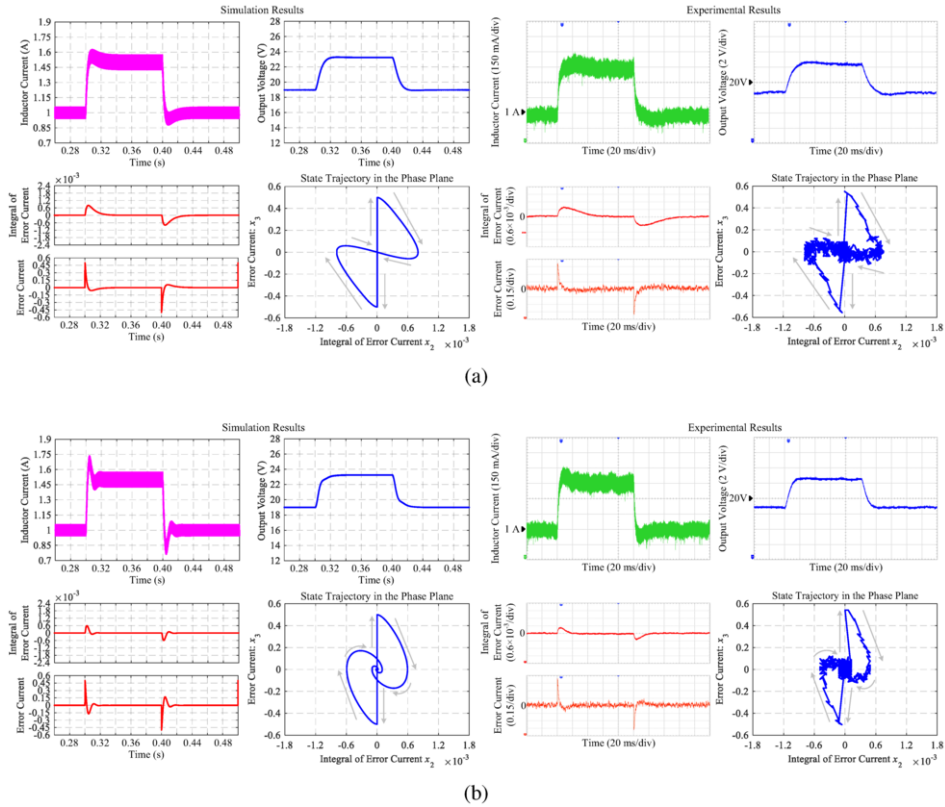


Figure 5. Inductor current, output voltage, current error, and the integral of the current error for DISMC in the case of $K_p = 1$ with (a) $K_i = 100$ and (b) $K_i = 500$.

computation of the equivalent control signal, which includes a significant integral error term, the performance of steady-state error rejection is consequently enhanced. Simulation and experimental results were utilized to confirm the hypothesis and the proposed control scheme. The inductor current can effectively track its reference, even in the presence of a sudden change in the current reference and variations in output voltage. The trajectories of the current error always converged towards zero.

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